

Srija Karicherla

Data Engineer | Python, SQL, BigQuery & Streaming

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Professional Summary

Data Engineer with **6+ years of experience** building, operating, and troubleshooting production data pipelines for high-volume event, transaction, telemetry, and business-system data. Strong hands-on experience with **Python, SQL, BigQuery, Google Cloud Platform, Kafka, Pub/Sub, Cloud Run, Cloud Storage, Airflow, dbt, Docker, GitHub Actions, and CI/CD**. Experienced designing partitioned and clustered warehouse tables, incremental pipelines, streaming consumers, data-quality checks, reconciliation jobs, and production monitoring. Skilled in debugging large-scale SQL workloads, reducing warehouse cost, improving pipeline reliability, and turning ambiguous stakeholder requirements into durable, documented datasets.

Technical Skills

Languages: Python, SQL, Bash, Java

Data Engineering: Batch Processing, Streaming Pipelines, ETL/ELT, Incremental Processing, Data Modeling, Data Quality, Schema Evolution, Reconciliation, Lineage, CDC Concepts

Google Cloud: BigQuery, Cloud Run, Pub/Sub, Cloud Storage, IAM, Cloud Logging, Cloud Monitoring, Cloud Composer

Streaming / Messaging: Apache Kafka, Pub/Sub, Consumer Groups, Partitioning, Offsets, Retries, Dead-Letter Queues, Event Replay, Idempotency

Warehouses / Databases: BigQuery, Snowflake, PostgreSQL, MySQL, ClickHouse, Partitioning, Clustering, Window Functions, Query Plans, Cost Optimization

Transformation / Orchestration: dbt, Apache Airflow, Cloud Composer, Scheduled Workflows, Dependency Management

DevOps / Software Practices: Docker, Kubernetes, Git, GitHub Actions, Jenkins, CI/CD, Terraform, Code Reviews, Automated Testing

Observability: Cloud Monitoring, Grafana, Prometheus, Structured Logging, Pipeline Metrics, Alerting, Data Freshness Monitoring

Professional Experience

Cloudflare

Senior Data Engineer

Jul 2025 – Present

Remote, USA

Technologies: Python, SQL, BigQuery, Kafka, Pub/Sub, Cloud Run, Cloud Storage, dbt, Airflow, ClickHouse, Docker, Kubernetes, GitHub Actions, Terraform

- Build and operate Python ingestion pipelines that process high-volume network and application telemetry, normalizing raw events into analytics-ready datasets used by engineering and operational teams.
- Develop Kafka and Pub/Sub consumers that validate, enrich, and route event data by schema and event type, with explicit handling for malformed records, retries, dead-letter queues, and replay.
- Deploy Python ingestion and transformation services to Cloud Run using containerized builds, environment-specific configuration, service accounts, IAM roles, and automated GitHub Actions deployments.
- Design BigQuery tables for large append-heavy datasets using date and event-time partitioning, clustering on frequently filtered identifiers, and incremental processing to avoid full-table scans.
- Write complex SQL using joins, CTEs, window functions, arrays, aggregations, and incremental merge patterns to transform raw event streams into well-defined analytical models.
- Review BigQuery execution details and bytes scanned to identify expensive joins, poor partition pruning, unnecessary column reads, and repeated transformations.
- Build dbt models and tests for event-level, session-level, and aggregate datasets, including uniqueness, null, accepted-value, relationship, and freshness checks.
- Develop Airflow workflows for scheduled ingestion, transformations, backfills, data-quality validation, and downstream dataset publication, with explicit retry and failure callbacks.
- Implement reconciliation jobs in Python and SQL to compare source-event counts, ingested records, transformed rows, and downstream aggregates, surfacing missing or duplicated data.
- Instrument pipelines with metrics for throughput, processing lag, failed records, data freshness, query duration, and service errors, with alerts for sustained deviations.
- Troubleshoot production pipeline failures by tracing event identifiers from Kafka/Pub/Sub through Cloud Run logs, BigQuery staging tables, transformation jobs, and downstream models.

- Reduced BigQuery processing cost by approximately **28%** by improving partition pruning, clustering high-cardinality access paths, and converting repeated full refreshes to incremental transformations.
- Improved a high-volume telemetry pipeline's end-to-end processing time by approximately **24%** through larger batch writes, parallel transformation stages, and removal of redundant enrichment queries.

PayPal

Data Engineer

Nov 2023 – May 2025

Remote, USA

Technologies: Python, SQL, Kafka, Snowflake, PostgreSQL, Airflow, dbt, Docker, AWS, Git, Jenkins

- Developed Python and SQL pipelines for high-volume payment and transaction events, producing curated datasets used for operations, reconciliation, risk analysis, and reporting.
- Built Kafka consumers for transaction events with schema validation, consumer-group processing, retry handling, dead-letter routing, and duplicate-event protection.
- Created incremental warehouse models for transaction, account, workflow, and settlement data, using event timestamps and business keys instead of repeatedly rebuilding full histories.
- Wrote SQL transformations involving multi-table joins, window functions, deduplication, conditional aggregation, and late-arriving transaction handling.
- Developed Python reconciliation jobs to compare source transaction counts and amounts against processed warehouse records, flagging missing, duplicated, or mismatched transactions for investigation.
- Built Airflow DAGs for ingestion, validation, transformation, and publication workflows, with dependencies, retries, SLAs, backfill support, and failure notifications.
- Added dbt tests and pipeline checks for uniqueness, nullability, referential integrity, freshness, row-count deviations, and financial reconciliation.
- Reduced processing time for a high-volume transaction dataset by approximately **25%** after converting repeated full loads to incremental processing and optimizing large SQL joins.

Deloitte

Data Engineer

May 2020 – Jul 2023

Hyderabad, India

Technologies: Python, SQL, PostgreSQL, MySQL, Kafka, Airflow, AWS, S3, Docker, Jenkins, Git

- Developed Python and SQL pipelines to ingest data from REST APIs, relational databases, CSV/JSON files, and enterprise source systems into analytical data stores.
- Built reusable Python modules for file validation, schema checks, type conversion, deduplication, data cleansing, and transformation.
- Created SQL transformations using joins, CTEs, window functions, aggregations, and business rules to produce reporting and operational datasets.
- Developed incremental ingestion logic using timestamps and source-system keys to process only new or changed records rather than repeatedly reloading full datasets.
- Built Airflow workflows for daily ingestion, transformation, validation, and delivery jobs, with dependencies, retry handling, logging, and failure notifications.
- Implemented data-quality controls for missing values, duplicate records, invalid reference values, row-count variances, and source-to-target reconciliation.
- Worked with business analysts and client teams to translate reporting and operational requirements into source mappings, transformation rules, dataset definitions, and validation criteria.
- Reduced manual data-validation effort by approximately **26%** by automating source-to-target reconciliation, row-count checks, and exception reporting inside scheduled pipelines.

Education

Montclair State University

Master of Science in Computer Science

Aug 2023 – May 2025

New Jersey, USA

Certifications

- **AWS Certified Data Engineer – Associate**
- **Google Data Analytics Professional Certificate**